

CS 696A – Full-Stack Enterprise Application Development – Prof. Radiy Matveev

- Create a file named “CS696A_WK7_{FirstName}_{LastName}.docx” (e.g., CS696A_WK7_Doosan_Baik.docx).
- Following through two (2) exercises below, answer eight (8) questions accompanied inside the file created.
- Upload the file to the assignment section of the course homepage at <https://classes.pace.edu> before the class end time.

Synopsis #1: Course Management System

You are designing a course management system for the university (like Pace Classes). See below for the functional requirements:

- An instructor manages multiple course sections and each course section have its own homepage.
- An instructor uploads weekly course materials and assignments.
- An instructor publishes the scores for the students.
- An instructor publishes the final grades for the students.
- A student sees all course sections they are enrolled in.
- A student sees weekly course materials and assignments as soon as the instructor uploads them.
- A student sees his/her own scores and final grades as soon as the instructor publishes them.

Question 1: The functional requirements above are intentionally kept ambiguous. What other functional requirements can you add to add to make the requirements crisper? Suggest 2 – 3 additional functional requirements.

Question 2: Consider CAP Theorem for non-functional requirements. What two (2) properties out of consistency, availability, partition tolerance must be defined? Explain why.

Question 3: What data would you have to store to support the requirements? Explain all.

Question 4: What API would you have to define to support the requirements? Explain all.

Synopsis #2: Photo Sharing Website

You are designing a website where you can share the photographs you’ve taken and other people can browse all the photos they are interested in (like Flickr, Pinterest). See below for the functional requirements:

- An uploader uploads the photographs he/she took with the captions and hashtags.
- An uploader modifies or deletes the captions and hashtags for the photographs he/she uploaded.
- A browser sees the list of photos he/she is interested based in the hashtags.
- A browser comments and likes the photos.
- A browser shares the links to the photos with the people he/she would like to share with.
- Anyone reports the photographs that are deemed inappropriate.
- An administrator deletes the photographs that are reported after the review.

Question 5: The functional requirements above are intentionally kept ambiguous. What other functional requirements can you add to add to make the requirements crisper? Suggest 2 – 3 additional functional requirements.

Question 6: Consider CAP Theorem for non-functional requirements. What two (2) properties out of consistency, availability, partition tolerance must be defined? Explain why.

Question 7: What data would you have to store to support the requirements? Explain all.

Question 8: What API would you have to define to support the requirements? Explain all.